MINI-PROJECT

TITLE:SMART STREET LIGHTING SYSTEM

Batch-16

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ABSTRACT

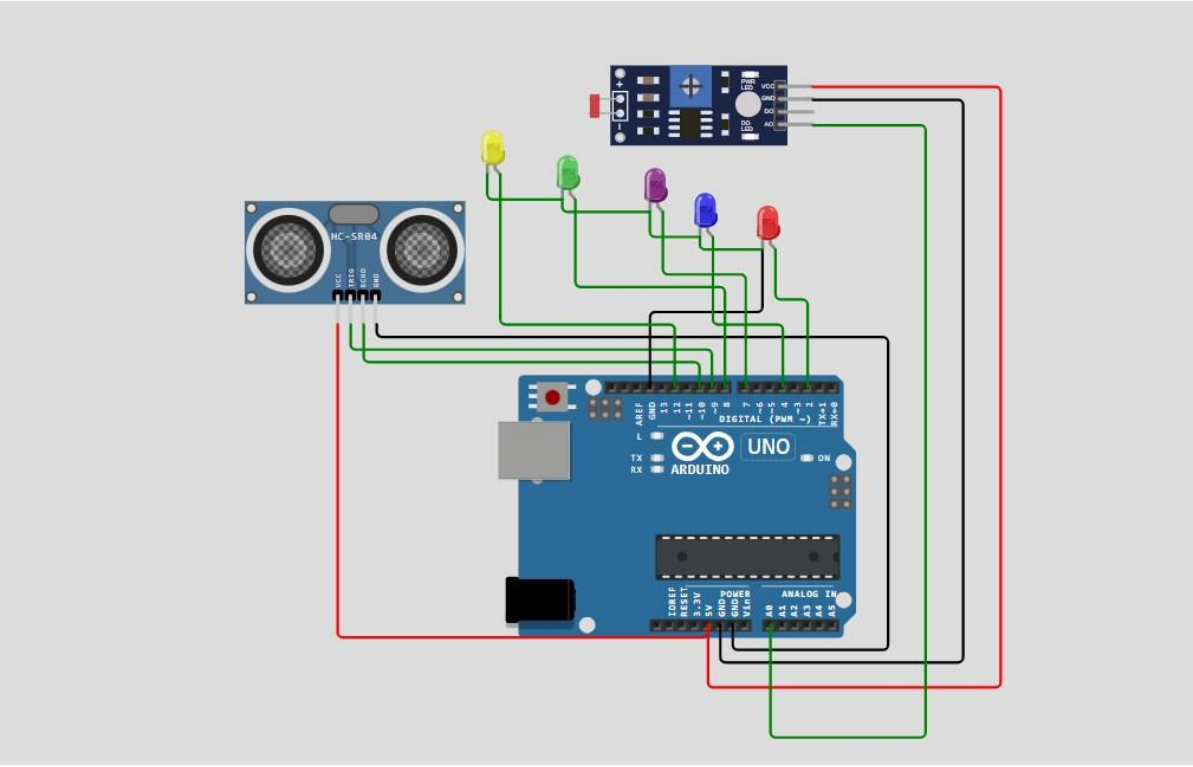
The development of smart street lighting systems is a significant advancement in energy conservation and urban management, driven by the need for sustainability and efficiency in modern cities. This paper presents a design for a smart street lighting system that leverages both ultrasonic sensors and Light Dependent Resistor (LDR) sensors to optimize the operation of street lights based on real-time environmental conditions. The system uses ultrasonic sensors for detecting the presence of vehicles and pedestrians, enabling streetlights to automatically adjust their brightness based on traffic flow. Meanwhile, LDR sensors monitor ambient light levels, ensuring that the streetlights are only active during low-light conditions, thus conserving energy during daylight hours.

The proposed system employs an intelligent control unit that processes data from the sensors to dynamically manage lighting. When no movement is detected by the ultrasonic sensors, the system reduces the lighting intensity or switches the lights off entirely, while during peak traffic hours, the lights increase in brightness for improved visibility and safety. Furthermore, the LDR sensor adjusts lighting levels based on natural sunlight, ensuring that the streetlights remain off during daylight and are activated when the ambient light falls below a certain threshold.

INTRODUCTION

As cities continue to grow and evolve, the need for smarter, more sustainable infrastructure becomes increasingly urgent. One of the most energy-intensive aspects of urban environments is street lighting, which traditionally operates regardless of external conditions, leading to unnecessary energy consumption and increased costs. In response to this challenge, the concept of smart street lighting has gained significant attention. By integrating advanced sensor technologies, such as ultrasonic and Light Dependent Resistor (LDR) sensors, street lighting systems can be made more responsive and efficient. Ultrasonic sensors detect the presence of vehicles and pedestrians, allowing for dynamic adjustment of light intensity based on real-time traffic flow, while LDR sensors monitor ambient light levels to ensure that lights are only on when needed. This intelligent system not only conserves energy but also improves safety and reduces maintenance costs, offering a sustainable solution for modern cities.

BLOCK DIAGRAM



COMPONENTS REQUIRED

- Arduino Uno
- Ultra sonic sensor
- LDR(LightDependent Resistor)
- LED(forlightcontrolsimulation)
- Jumper Wires.
- Resistors(10kΩforLDRvoltage divider)
- BreadboardandJumperWires

WORKING PRINCIPLE

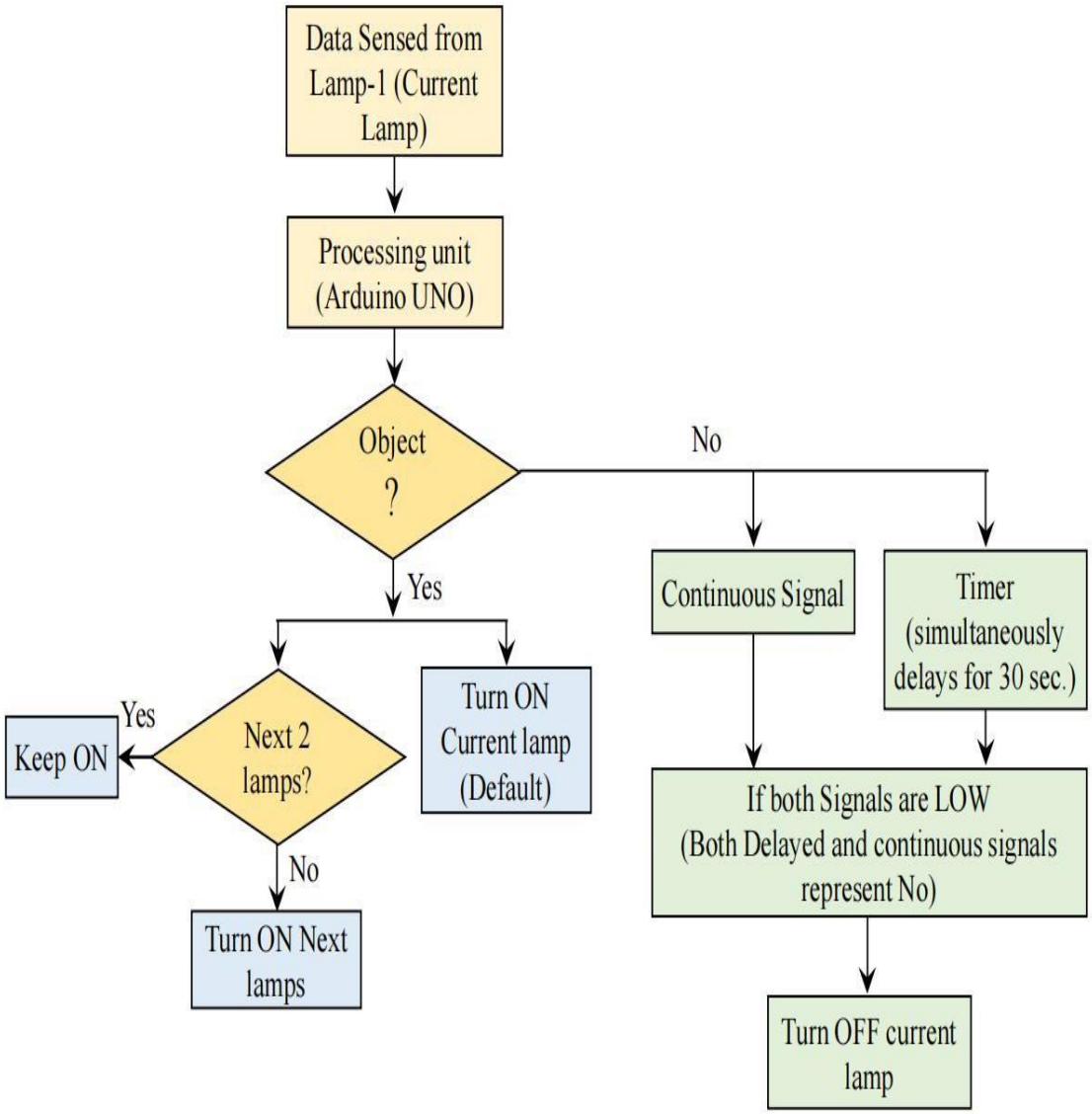
Ultrasonic Sensor Integration: Ultrasonic sensors are strategically placed on streetlights to detect the presence of vehicles or pedestrians. These sensors use sound waves to measure the distance between the sensor and an object. When the sensor detects motion, it sends a signal to the control unit, triggering the streetlight to increase its brightness to provide better illumination for safety and visibility. Once the area becomes clear, the sensor signals the system to reduce the light intensity or turn it off, conserving energy when no activity is detected.

LDR Sensor Integration: The Light Dependent Resistor (LDR) sensor is used to measure ambient light levels. During daylight hours, when there is enough natural sunlight, the LDR sensor detects the high light intensity and sends a signal to the control unit to keep the streetlights turned off, thus saving energy. As dusk approaches and ambient light decreases, the LDR sensor detects lower light levels and automatically turns the streetlights on, ensuring they provide adequate lighting as needed.

Intelligent Control Unit: The control unit processes the data received from both the ultrasonic and LDR sensors. It adjusts the lighting intensity based on the sensor inputs, optimizing energy use while ensuring adequate lighting. The system is programmed to activate the lights when there is low ambient light or movement detected, and to turn them off or dim them when conditions do not require illumination.

Energy Conservation: By dynamically adjusting lighting based on traffic and ambient light levels, the system minimizes energy consumption. The streetlights are only on when necessary, providing full illumination during peak hours or when movement is detected, and reducing brightness or turning off completely when no movement or sufficient daylight is present.

ALGORITHM


Data Sensed from Lamp-1 (Current Lamp)
Processing unit (Arduino UNO)
Object ?
No
Yes
Continuous Signal
Timer (simultaneously delays for 30 sec.)
Next 2 lamps?
Yes
Keep ON
Turn ON Current lamp (Default)
No
Turn ON Next lamps
If both Signals are LOW (Both Delayed and continuous signals represent No)
Turn OFF current lamp

CODE

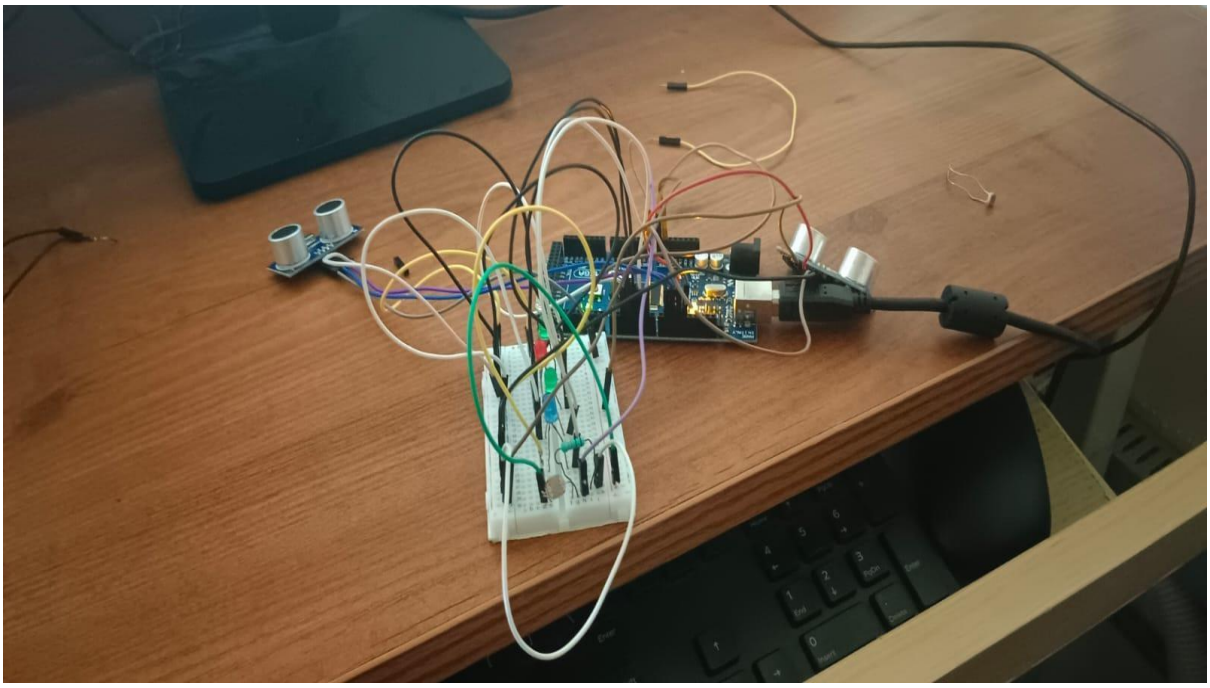
```
const int LED1_PIN = 2;
const int LED2_PIN = 3;
const int LED3_PIN = 4;
const int LED4_PIN = 5;
const int LED5_PIN = 6;

const int TRIG1_PIN = 9;
const int ECHO1_PIN = 10;
const int TRIG2_PIN = 11;
const int ECHO2_PIN = 12;

const int LDR_PIN = A0;  // LDR pin connected to A0 (Analog input)

// Define the threshold for light intensity (LDR value)
const int LIGHT_THRESHOLD = 800;  // You may need to adjust this based on your setup

// Define the distance threshold for object detection (in cm)
const int DISTANCE_THRESHOLD = 100;  // Turn on light if object is within 100 cm

void setup() {
  // Set LED pins as output
  pinMode(LED1_PIN, OUTPUT);
  pinMode(LED2_PIN, OUTPUT);
  pinMode(LED3_PIN, OUTPUT);
  pinMode(LED4_PIN, OUTPUT);
  pinMode(LED5_PIN,  OUTPUT);

  // Set Ultrasonic Trigger pins as output and Echo pins as input
  pinMode(TRIG1_PIN, OUTPUT);
  pinMode(ECHO1_PIN, INPUT);
  pinMode(TRIG2_PIN, OUTPUT);
  pinMode(ECHO2_PIN, INPUT);
```

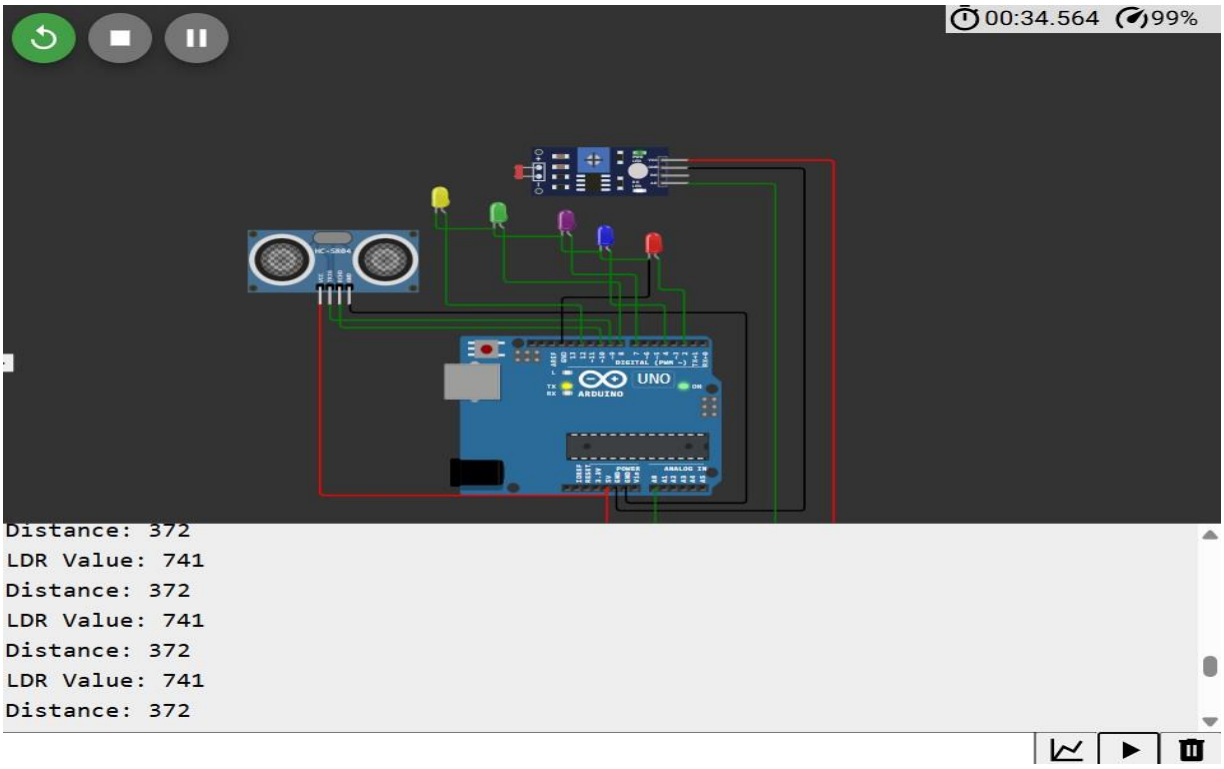
```
// Start serial communication
  Serial.begin(9600);
}

void loop() {
  // Measure distance for both ultrasonic sensors
  long distance1 = measureDistance(TRIG1_PIN, ECHO1_PIN);
  long distance2 = measureDistance(TRIG2_PIN, ECHO2_PIN);

  // Read the LDR value (light intensity)
  int ldrValue = analogRead(LDR_PIN);
  Serial.print("LDR Value: ");
  Serial.println(ldrValue);
  Serial.print("Distance Sensor 1: ");
  Serial.print(distance1);
  Serial.println(" cm");
  Serial.print("Distance Sensor 2: ");
  Serial.print(distance2);
  Serial.println(" cm");

  // Handle LED behavior for Ultrasonic Sensor 1 (3 LEDs)
  controlLEDsForSensor(ldrValue, distance1, LED1_PIN, LED2_PIN, LED3_PIN);

  // Handle LED behavior for Ultrasonic Sensor 2 (2 LEDs)
  controlLEDsForSensor(ldrValue, distance2, LED4_PIN, LED5_PIN);

  delay(1000);  // Delay for 1 second before the next loop
}

// Function to measure distance using ultrasonic sensor
long measureDistance(int trigPin, int echoPin) {
  // Send a 10us pulse to trigger the ultrasonic sensor
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
```

```
 digitalWrite(trigPin, LOW);

  // Read the pulse duration from the Echo pin
  long duration = pulseIn(echoPin, HIGH);

  // Calculate the distance in cm
  long distance = duration * 0.0344 / 2;  // Speed of sound is 0.0344 cm/us, and we divide by 2
for

round trip
  return distance;
}

// Function to control LEDs for each ultrasonic sensor
void controlLEDsForSensor(int ldrValue, long distance, int led1, int led2, int led3) {
  // Check if it's dark (LDR value < LIGHT_THRESHOLD)
  if (ldrValue < LIGHT_THRESHOLD) {
    // If an object is detected (distance < DISTANCE_THRESHOLD), turn on the LEDs
    if (distance < DISTANCE_THRESHOLD) {
      digitalWrite(led1,   HIGH);
      digitalWrite(led2,   HIGH);
      digitalWrite(led3,  HIGH);
      Serial.println("Object detected and it's dark: LEDs ON");
    }
    else {
      // If no object is detected (distance >= DISTANCE_THRESHOLD), turn off the LEDs
      digitalWrite(led1, LOW);
      digitalWrite(led2, LOW);
      digitalWrite(led3,  LOW);
     Serial.println("No object detected and it's bright: LEDs OFF");
    }
  }
}

// Overloaded function for 2 LEDs (for Sensor 2)
void controlLEDsForSensor(int ldrValue, long distance, int led1, int led2) {
```

```
  if (ldrValue < LIGHT_THRESHOLD) {
    // If an object is detected (distance < DISTANCE_THRESHOLD), turn on the LEDs
    if (distance < DISTANCE_THRESHOLD) {
      digitalWrite(led1, HIGH);
      digitalWrite(led2, HIGH);
      Serial.println("Object detected and it's dark: LEDs ON");
    }
    else {
      // If no object is detected (distance >= DISTANCE_THRESHOLD), turn off the LEDs
      digitalWrite(led1, LOW);
      digitalWrite(led2, LOW);
      Serial.println("No object detected and it's dark: LEDs OFF");
    }
  }
  // If it's bright (LDR value >= LIGHT_THRESHOLD)
  else {
    // If an object is detected (distance < DISTANCE_THRESHOLD), turn off the LEDs
    if (distance < DISTANCE_THRESHOLD) {
      digitalWrite(led1, LOW);
      digitalWrite(led2, LOW);
      Serial.println("Object detected but it's bright: LEDs OFF");
    }
    else {
      // If no object is detected (distance >= DISTANCE_THRESHOLD), turn off the LEDs
      digitalWrite(led1, LOW);
      digitalWrite(led2, LOW);
      Serial.println("No object detected and it's bright: LEDs OFF");
    }
  }
}
```

OUTPUT




00:34.564
99%
UNO
ARDUINO
Distance: 372
LDR Value: 741
Distance: 372
LDR Value: 741
Distance: 372
LDR Value: 741
Distance: 372

ADVANTAGES

- Wastage of electrical power for enlightening streets has been considerably minimized in comparison to most efficient HPS lamps.
- Energy-efficient as lights operate only when needed.
- Reduces electricity costs and power consumption.
- Increases lifespan of street lights by reducing continuous operation.
- Enhances road safety by ensuring proper lighting when required.

DISADVANTEGES

- Initial installation cost may be high.
- Sensors may malfunction due to environmental conditions (rain, dust, etc.).
- Requires periodic maintenance to ensure sensor accuracy.
- Cybersecurity Risks
- Privacy Concerns

FUTURE SCOPE

1. Integration with IoT and Cloud

- Connecting the system to the Internet of Things (IoT) can enable remote monitoring and control of streetlights.
- Cloud-based data storage can help analyze lighting patterns and energy consumption over time.

2. Adaptive Lighting for Energy Efficiency

- AI-based optimization can adjust brightness based on real-time traffic density and weather conditions.
- Dynamic control of brightness levels can reduce energy wastage while ensuring road safety.

3. Solar-Powered Street Lights

- Integration with solar panels can make the system self-sustainable, reducing reliance on grid power.
- Battery storage can be used to operate lights during the night.

4. Smart Traffic Management

- Ultrasonic sensors can help analyze traffic patterns and adjust streetlight intensity accordingly.
- Can be integrated with smart traffic lights to improve road safety and reduce congestion.

5. Fault Detection and Maintenance Alerts

- The system can detect faulty or non-functioning streetlights and send alerts for maintenance and repair.
- Predictive maintenance can reduce the downtime of streetligh

APPLICATIONS

- Traffic & Pedestrian Safety
- Smart City Integration
- Environmental Monitoring
- Security & Surveillance
- IoT & Automation
- Disaster Management & Emergency Services

CONCLUSION

The proposed ultrasonic sensor-based energy-efficient streetlight system has been demonstrated to reduce consumption of energy. A prototype has been designed in the lab scale, its hardware implementation and effectiveness have been vindicated. This paper has explained its efficiency and energy-sving aspects considering the most advanced HPS and LED lamps developed by ‘Philips’. The use of an ultrasonic sensor and Arduino processor to control automatic street lighting system at night has been successfully performed. The intelligent operation of the system may stop working as soon as the sensor becomes spoiled. This system functions in one way traffic which can further be extended suitable for two-way traffic. GSM based service can be added to design a smart control of the street lighting system.

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- M. Srikanth, and K. N. Sudhakar, “Zigbee based remote control automatic street light system,” Int. J. Eng. Sci. Comput., pp. 639-643, 2014.